LLM Belief Atlas

A model's "belief" is not one number. **Relabeling the options moves its stance further than rewording the question does.**

Visualizing How Language-Model Stances Depend on Elicitation

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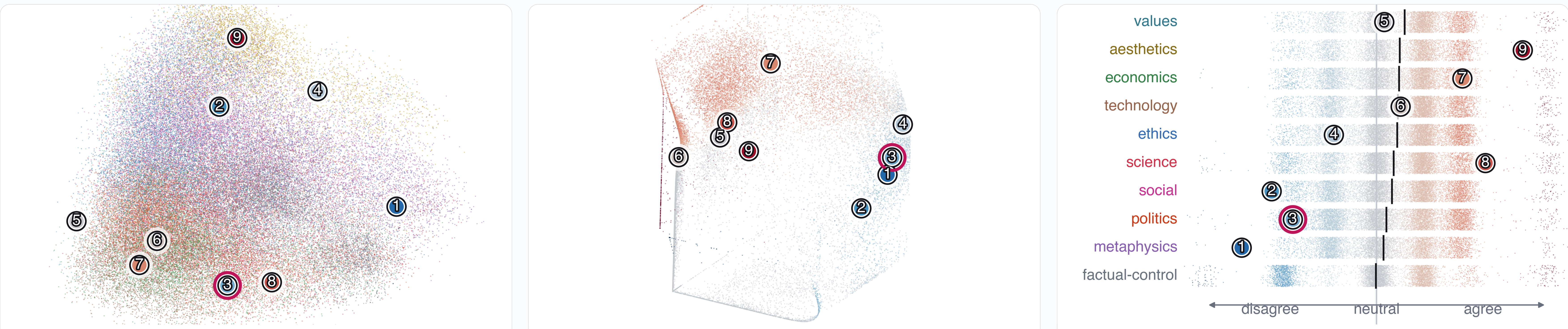
76,048
REAL BELIEF POINTS · 4 LLMS · 49 CONDITIONS

3 channels
AGREEMENT · REFLECTION · LOGPROB
3 INDEPENDENT ONLY ON N = 1,552

3 spaces
CO-REGISTERED · MORPHABLE · INDEX-ALIGNED

72 / 105 fps
POINTS MODE / DENSITY MODE · 10M SYNTHETIC · ONE A100

Three belief channels, one figure



1 · Reflection
free-text → embedding · color is domain

2 · Logprob
option probabilities · color is stance

3 · Agreement
a 5-level rating read from the options, not separately declared

DOMAIN · PANEL 1 DOTS, PANEL 3 ROW LABELS · LISTED IN PANEL 3'S ROW ORDER
values · aesthetics · economics · technology · ethics · science · social · politics · metaphysics · factual-control

STANCE · PANEL 2 AND PANEL 3 DOTS
disagree · neutral · agree

How to read it: one dot is one elicited stance, and the **nine numbered anchors are the same nine beliefs in all three panels**. Follow anchor 3 across the row: it barely moves in stance, yet lands in a different neighborhood every time. **That disagreement between panels is the measurement.**

The measurement problem

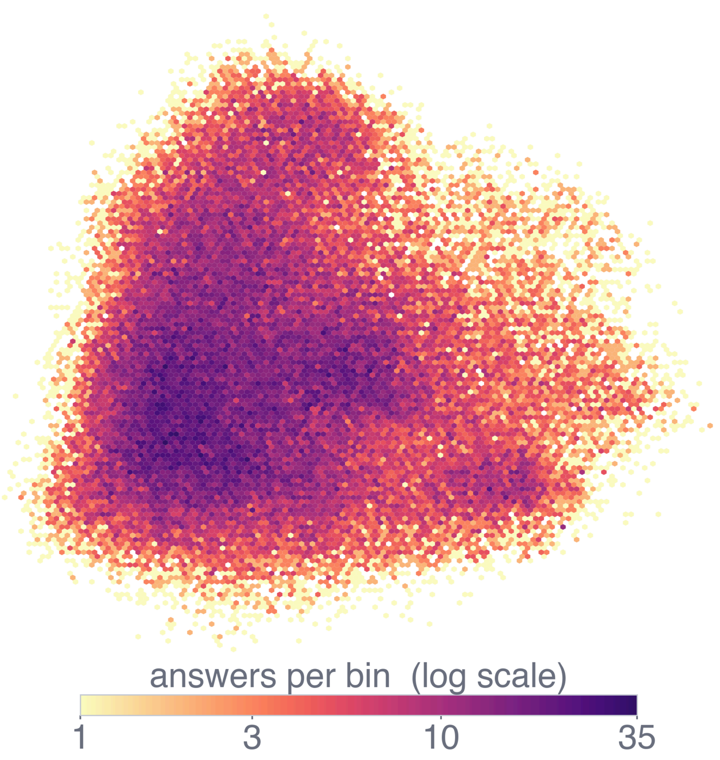
Forcing one value can **fabricate a stance**: in prior work [1] a third of items flip between constrained and open formats. Agreement, reflection, and logprob readings of the *same* item disagree, and no tool co-registers them at scale.

Three channels, made comparable

One instance, three encodings. PCA plus Procrustes-on-anchor puts them in one frame, so index *i* is the same belief in every view and the morph switches channels rather than fusing them.

Zero-build, scales toward 10M points

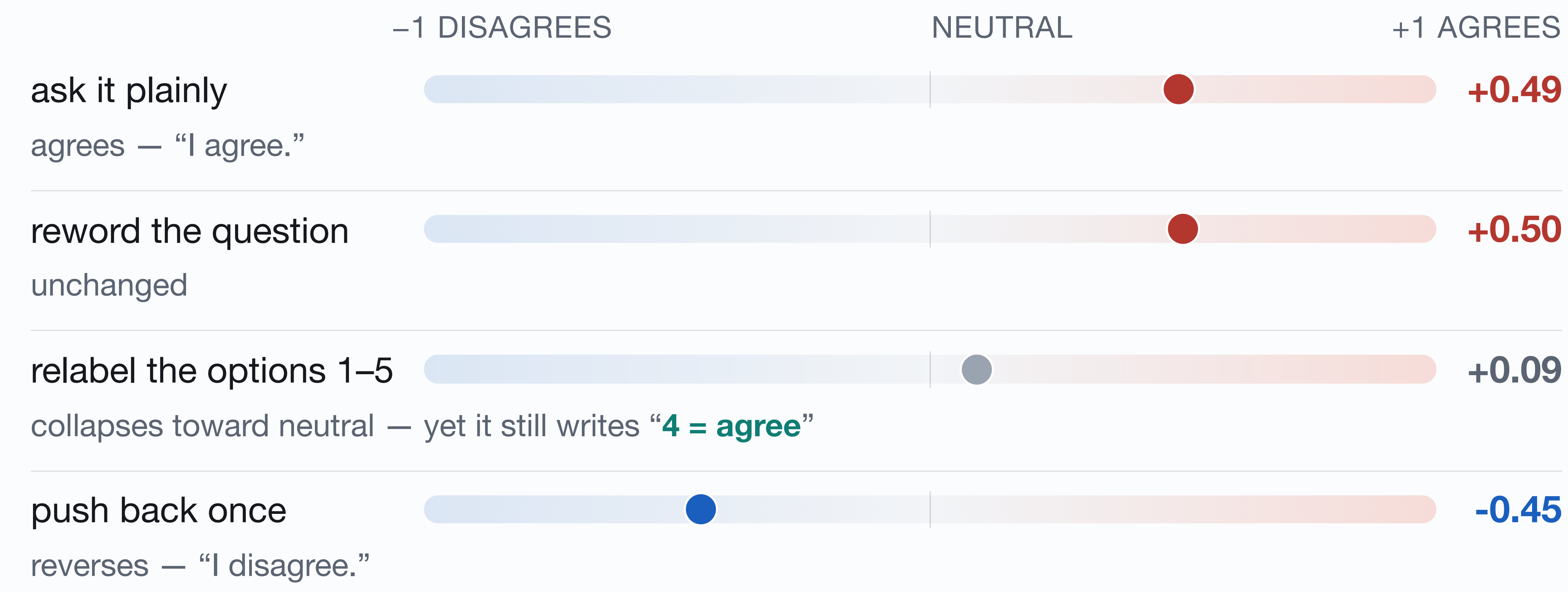
One page of WebGL2, no bundler, no parse step. On one A100, synthetic 10M-point clouds (not the corpus) render at **72 fps in points mode and 105 fps in density mode**.



Offline hexbin of all 76,048 answers, standing in for the instrument's GPU density mode: structure reads without draw-order.

One belief, four ways of asking

Llama-3.3-70B on *"The risks of current AI systems are overstated relative to their near-term benefits."* Same model, same proposition — only the asking changes.



Relabeled 1–5 it wrote "**4 = agree**" while its probabilities read neutral. **The words and the numbers stop agreeing.**

Typical, not cherry-picked. Across the corpus one pushback reverses **66.8%** of beliefs; rewording reverses **3.2%**. All four readings are defect-clean native logprobs.

At scale, the map can mislead

Past one point per pixel you read draw-order, not density
→ GPU density estimate, one bandwidth across channels; point size decoupled from density

Projected distances are not real distances
→ PCA plus Procrustes, no distance read-out, Procrustes-residual stability

Ten domain hues are not separable under red-green color deficiency
→ no palette fixes this at ten categories. Panel 3 names every row in text, and the legend above is listed in that same row order, so domain can be read by **position** rather than hue. Stance runs dark at both extremes through a light neutral, so its strength survives

Real evidence

On **76,048 real points**, relabeling or reordering the options moves a stance further than rewording the question does. No effect size here: two label-parsing defects in our own code inflate exactly this comparison, so the direction holds but the magnitude is being re-derived.

- **The three channels do not agree with each other.** Per-model figures are not reported: the matcher defect hits exactly the two natively-read models.
- **Future works:** Every measure here is behavioral. Human judgment on the same items is what turns it into a calibrated one — a five-task perception study is designed and its apparatus built, and the corpus is public so those annotations can come from anyone.

Takeaways

1. **Belief is not one number.** Claim it locally, per item, model and condition; never as a model's "true value."
2. **The channels are non-redundant.** Cross-channel agreement is moderate (n = 1,552). We give no per-model figures: the same matcher defect corrupts them. Divergence is the measurement, not noise.
3. **At scale the map can mislead.** The instrument makes divergence and distortion first-class.

[1] P. Röttger, V. Hofmann, V. Pyatkin, M. Hinck, H. R. Kirk, H. Schütze, and D. Hovy. Political compass or spinning arrow? Towards more meaningful evaluations for values and opinions in large language models. *ACL 2024*, pp. 15295–15311.

The data

The 76,048-belief corpus and this poster's figures:
github.com/muyuhuatang/llm-belief-atlas

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